

Joseph Peters

Aspiring Engineer & Innovator

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Summary & Profile

Aspiring engineer passionate about **robotics, game development, and building real-world technology** that addresses critical societal and environmental challenges. Driven to solve problems that matter through creativity, collaboration, and code. Proven leader in technical and philanthropic initiatives, with a strong foundation in programming and engineering tools.

Education

St Andrews Episcopal School | Potomac, MD

- Expected Graduation: 2026

Johns Hopkins University | Baltimore, MD

- **Engineering Innovation Program** (Summer 2025)
 - Completed a competitive, college-level course in core engineering disciplines (mechanical, electrical, structural, computer).
 - Designed, tested, and presented real-world solutions to multi-week engineering challenges. Achieved an **A** in the course.

Montgomery College | Rockville, MD

- **Intermediate Java Programming Certificate** (Summer 2024)
 - Four-week college-level course covering object-oriented programming fundamentals, including classes and control structures.

Featured Projects & Technical Experience

DiscoverU - AI Career Exploration Platform

Founder & Lead Developer | 2023-Present

- **First Place Winner** at the 2025 HackBac Competition for designing an innovative career exploration platform.
- Developed a gamified simulation platform to bridge the gap between education and real-world careers, guiding students through virtual workplace scenarios.
- Authored and published a research paper comparing the DiscoverU model to traditional counseling methods after conducting user feedback research across multiple schools.

Trash-Sorting Robot

Self-Initiated Engineering Project | 2024-Present

- Developing a robotics system utilizing **computer vision** for real-time identification and sorting of waste materials to reduce recycling contamination.
- Focusing on object detection models, hardware configuration, and ensuring the system meets specific sustainability metrics for environmental impact.

Robotics Club

Programming & Design Lead | St Andrews Episcopal School | 2022-Present

- Led programming and design for competition-ready robots, utilizing **CAD (Fusion 360)** and hands-on fabrication.
- Competed in regional tournaments, demonstrating technical skills in system optimization and troubleshooting.

Tech Internship

Nika Corporation | July-August 2024

- Shadowed engineers and observed real-world engineering processes within a government contracting environment.
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Leadership & Community Impact

GiveBack Backpacks Foundation (givebackpacks.co)

Co-Founder | 2022-Present

- **Raised over \$70,000** to provide laptop-equipped backpacks to students in underserved communities, directly contributing to bridging the digital divide.
- Led international distribution efforts, delivering technology and supplies to students in Ghana.

Leukemia & Lymphoma Society (LLS)

Team Captain | 2024

- **Raised over \$50,000** in 7 weeks for cancer research, finishing as 2nd Runner-Up for DC Student Visionaries Team Member of the Year (3rd out of 400+ students).
- Spearheaded school-wide fundraising, donor outreach, and advocacy strategy.

Chess Club - “Lions”

Co-Captain | St Andrews Episcopal School | 2022-Present

- Co-led the school chess team and organized inter-school competitions.
 - Prototyped a web-based platform to connect school chess clubs for remote tournaments.
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Technical Skills

Category	Skills
Programming	Python, Java, JavaScript, C++, HTML/CSS, React.js
Engineering Tools	Arduino, CAD (Fusion 360), Circuit Design
Development	Git, Web Development, Prototyping, System Optimization

Additional Experience

- **HackBac Competition (Team Member, 2022-Present):** Developed prototypes addressing community challenges, including mental health and gentrification.
- **Achieve Balance App (2022):** Designed and presented a mental health and scheduling app prototype specifically for athletes, focusing on wellness and accessibility.